

Trajectory-Based Vulnerability Response

A GyroAuth Applied Model for Post-Login Security Monitoring

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Repository: <https://github.com/gitGyro-Dev/gyroauth>

Applied PoC: examples/trajectory_vulnerability_response/

Related GyroAuth preprint: <https://doi.org/10.51094/jxiv.4600>

Abstract

As AI-assisted vulnerability discovery advances, security models that rely solely on preventing or patching known vulnerabilities may become insufficient. This paper proposes **Trajectory-Based Vulnerability Response** as an applied GyroAuth model for post-login security monitoring. The central claim is that an exploit may consist of individually valid events while forming an impermissible and unstable operational trajectory. In this model, a vulnerability is interpreted as a structural weakness that allows an impermissible trajectory to be accepted as normal execution. The proposed framework consists of three layers: **Boundary / Negative Definition** for prevention, **Trajectory Detection** for post-entry anomaly detection, and **Stability Response** for containment, re-authentication, isolation, and recovery. A minimal proof-of-concept reconstructs session trajectories from synthetic access logs, evaluates deviations in sequence, velocity, context, scope, volume, permission, and destination, computes a stability value from weighted deviation, and maps the resulting state to staged responses. In contrast to continuous authentication, risk-based or adaptive authentication, and behavioral biometric approaches, the proposed model is not primarily a user-identity classifier. It frames vulnerability response as the detection of unstable trajectories formed by otherwise valid events. This work does not attempt to discover vulnerabilities or reproduce exploits; rather, it demonstrates how GyroAuth can extend authentication from login-time identity verification to continuous trajectory convergence monitoring.

Keywords: GyroAuth; Gyro Logic; Trajectory Detection; Vulnerability Response; Post-Login Security; Stability Response; State Convergence; Cybersecurity; Authentication; Anomaly Detection; Continuous Authentication; Risk-Based Authentication; Behavioral Biometrics

1. Introduction

The increasing capability of AI systems to assist vulnerability discovery raises a practical question for security architecture. If unknown weaknesses can be found faster than they can be patched, security cannot rely only on the assumption that every vulnerability will be identified and closed before exploitation.

This does not imply that prevention, patching, input validation, and access control are unnecessary. They remain essential. However, modern systems also require a complementary layer: a method for detecting and responding to abnormal operational trajectories after an event has passed through normal processing paths.

Many attacks do not appear as obviously invalid events. A login may succeed. An API request may return 200 OK. A database query may complete. A file download may be allowed. Each event may be individually valid. The abnormality appears not in a single event but in the sequence, speed, scope, context, and destination of the events.

This paper therefore proposes the following core distinction:

Event validity is not trajectory stability.

Or, more compactly:

valid events != stable trajectory

This paper develops this distinction as a GyroAuth applied model called **Trajectory-Based Vulnerability Response**.

2. Background: GyroAuth

GyroAuth is an application-layer framework derived from Gyro Logic and positioned in relation to GyroOS. Its core definition is:

Authentication = Stability-based Selection over State Convergence

The GyroAuth framework has been published as a Jxiv preprint. This paper is an applied extension of that framework. While GyroAuth defines authentication as stability-based selection over state convergence, the present work applies the same framing to post-login vulnerability response.

In ordinary authentication systems, authentication is often treated as a point-in-time decision. A user provides credentials, a token is issued, and the session is treated as authenticated.

GyroAuth instead treats authentication as a state process. It observes whether a multi-dimensional state converges toward an expected identity trajectory, and then selects an authentication decision based on stability, deviation, operator response, and history.

This paper extends that idea from login-time identity verification to post-login session monitoring. The question is no longer only:

Is this user authenticated?

but also:

Does this post-login session trajectory remain stable within the expected identity, permission, and operational context?

This extension does not redefine Gyro Logic. It does not replace GyroOS. It is an applied GyroAuth note for post-login security monitoring.

3. Related Work and Positioning

This work is related to continuous authentication, risk-based authentication, adaptive authentication, behavioral biometrics, Zero Trust Architecture, and operational security monitoring. These areas provide important context for positioning the contribution of the present paper.

Continuous authentication addresses the limitation of one-time login by verifying a user's identity throughout an active session. Prior studies have explored continuous or active authentication using behavioral signals such as touch dynamics, keystroke dynamics, gait, motion patterns, voice, application usage, and other implicit features. Survey work on behavioral biometric authentication has shown that such approaches aim to verify or identify users implicitly and continuously, especially in mobile and sensor-rich environments.

Risk-based authentication and adaptive authentication take a complementary approach. Instead of applying the same challenge to every access attempt, they evaluate contextual and implicit signals such as device, location, network, time, or usage patterns, and trigger additional authentication factors when the risk level exceeds a threshold. Empirical studies have shown that risk-based authentication is used in real online services and is relevant for defending against credential stuffing, password leaks, and suspicious login attempts.

Behavioral biometric approaches, continuous authentication, and risk-based authentication are highly relevant to GyroAuth because they all move authentication beyond static credential matching. However, the focus of the present paper is different. This paper does not propose a new behavioral biometric classifier, does not evaluate biometric equal error rates, and does not primarily estimate whether the current operator is the legitimate user. Instead, it treats a post-login operation sequence as a trajectory and evaluates whether that trajectory remains stable within identity, permission, resource, volume, destination, and operational context.

The proposed model also differs from general anomaly detection and SIEM-style monitoring. Traditional monitoring may flag isolated events, known indicators, signatures, rules, or statistical anomalies. Trajectory-Based Vulnerability Response instead emphasizes the following claim:

Individually valid events may compose an impermissible trajectory.

The novelty of this work is therefore not the claim that continuous monitoring is useful. That is already established in continuous authentication, Zero Trust, and security monitoring. The novelty is the GyroAuth-based formulation of vulnerability response as trajectory stability evaluation: a vulnerability is interpreted as a structural weakness that allows an impermissible trajectory to be accepted as normal execution, and response is mapped through Stability Response rather than through event validity alone.

In this sense, the present paper is best positioned as a conceptual and implementation-oriented bridge between GyroAuth and existing post-login security monitoring. It complements continuous authentication and adaptive authentication by shifting the unit of analysis from user identity confidence alone to the stability of an operational trajectory.

4. Vulnerability as Impermissible Trajectory

A vulnerability is often described as a weakness in software, configuration, system design, or operation that can be exploited by an attacker. In the present model, the definition is sharpened in trajectory terms:

A vulnerability is a structural weakness that allows an impermissible trajectory to be accepted as normal execution.

This definition emphasizes that a vulnerability is not merely a bug. A bug is a failure to behave as expected. A vulnerability is a failure to preserve security boundaries, state-transition constraints, or operational assumptions.

Examples include:

- data being interpreted as executable command,
- an unauthenticated state entering an authenticated trajectory,
- a general user trajectory crossing into an administrative trajectory,
- a resource boundary allowing access to another user's data,
- a session continuing normally after abnormal data movement.

In GyroAuth terms, vulnerability response must therefore consider not only whether an event is accepted, but also whether the resulting trajectory remains stable.

5. Attack as Valid Events Forming an Unstable Trajectory

An attack does not always appear as a single invalid event. In many practical cases, the attacker uses valid operations in an abnormal sequence.

For example:

```
login  
-> api_access  
-> bulk_download  
-> external_transfer
```

Each event may be successful. The login may be valid. The API may be reachable. The download function may exist. The external transfer may be performed by an allowed endpoint. Yet the trajectory as a whole may be impermissible.

Let a session trajectory be defined as:

$$T_s = (e_1, e_2, \dots, e_n)$$

where each e_i is an event in session s .

Event validity may hold:

$$\forall e_i \in T_s, \text{allow}(e_i) = \text{true}$$

However, trajectory stability may still fail:

$$\text{Stable}(T_s \mid C_s) < \theta$$

where C_s denotes the context of the session, including user, role, device, time, location, resource, and operational history.

This is the central security claim of this paper:

All events may be allowed, but the trajectory may still be impermissible.

6. Three-Layer Model

The proposed model consists of three layers, as shown in Figure 1.

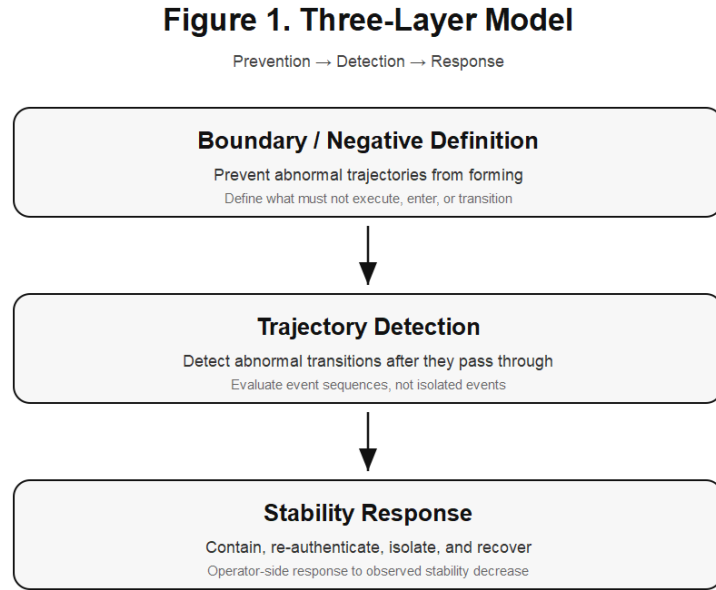


Figure 1: Three-Layer Model: prevention, detection, and response.

Trajectory-Based Vulnerability Response is structured into three layers:

Boundary / Negative Definition

Trajectory Detection

Stability Response

These correspond to prevention, detection, and response.

6.1 Boundary / Negative Definition

Boundary / Negative Definition is the preventive layer.

Security design is not only the definition of what a system can do. It also requires defining:

- what the system must not execute,
- which state transitions must not occur,
- which inputs must not enter command space,
- which roles must not cross resource boundaries,
- which uncertain states must not reach critical operations.

In this sense, Boundary / Negative Definition is a way to prevent abnormal trajectories from forming.

Examples include:

Unauthenticated -> Admin API : disallowed

User role -> All resources : disallowed

Input data -> SQL command space : disallowed

Unknown device -> Bulk export : disallowed or challenged

Unstable session -> External send : disallowed or restricted

This is prevention. It reduces the space in which an abnormal trajectory can be formed.

6.2 Trajectory Detection

Trajectory Detection is the post-entry detection layer.

Since no boundary is perfect, some abnormal trajectories may pass through ordinary controls. Trajectory Detection reconstructs event sequences from logs and evaluates whether they remain stable within the expected context.

Typical deviation categories include:

- sequence deviation,
- velocity deviation,
- context deviation,
- scope deviation,
- volume deviation,
- permission deviation,
- destination deviation.

The detection target is not only the event. It is the event sequence.

Detection target = trajectory, not isolated event

This layer is conceptually compatible with existing security monitoring practices, while emphasizing trajectory-level stability rather than isolated event validity.

6.3 Stability Response

Stability Response is the response layer.

Detection alone is not defense. When trajectory deviation is observed, the system must connect detection to staged control actions.

Possible responses include:

- continue,
- observe,
- challenge,
- restrict,
- isolate,
- recover.

In GyroAuth terms, Stability itself does not decide. Stability is an observed state variable. The operator-side response function maps observed stability and deviation to actions.

$$R = \rho(S, \Delta, C)$$

where S is observed stability, Delta is deviation, C is context, and rho is the operator-side response mapping.

7. Architecture for Existing Systems

The proposed model can be added to existing systems without immediately rewriting the application core. A minimal architecture is:

Existing Logs

- > Slice Normalizer
- > Trajectory Builder
- > Stability Evaluator
- > Deviation Detector
- > Response Controller

The overlay architecture is summarized in Figure 2.

The system first collects logs and events from the existing environment. These may include application logs, API gateway logs, authentication logs, authorization logs, database audit logs, file access logs, and session logs.

The Slice Normalizer converts these logs into a common event representation:

```
{
  "timestamp": "2026-05-21T10:15:30+09:00",
  "user_id": "user_123",
  "session_id": "sess_abc",
  "device_id": "dev_xyz",
  "source_ip": "203.0.113.10",
  "geo": "JP",
  "event_type": "api_access",
  "endpoint": "/api/export",
  "role": "user",
  "resource_type": "customer_data",
  "resource_id": "customers/*",
  "data_volume": 10485760,
  "status": "success"
}
```

The Trajectory Builder groups events by session, user, device, resource, or time window. The Stability Evaluator computes deviations from expected operational trajectories. The Response Controller maps the result to response actions through existing control points such as IAM, MFA, API Gateway, rate limiter, SIEM, SOAR, or session manager.

Figure 2. Existing-System Overlay Architecture

Logs → Slice → Trajectory → Deviation → Stability → Response

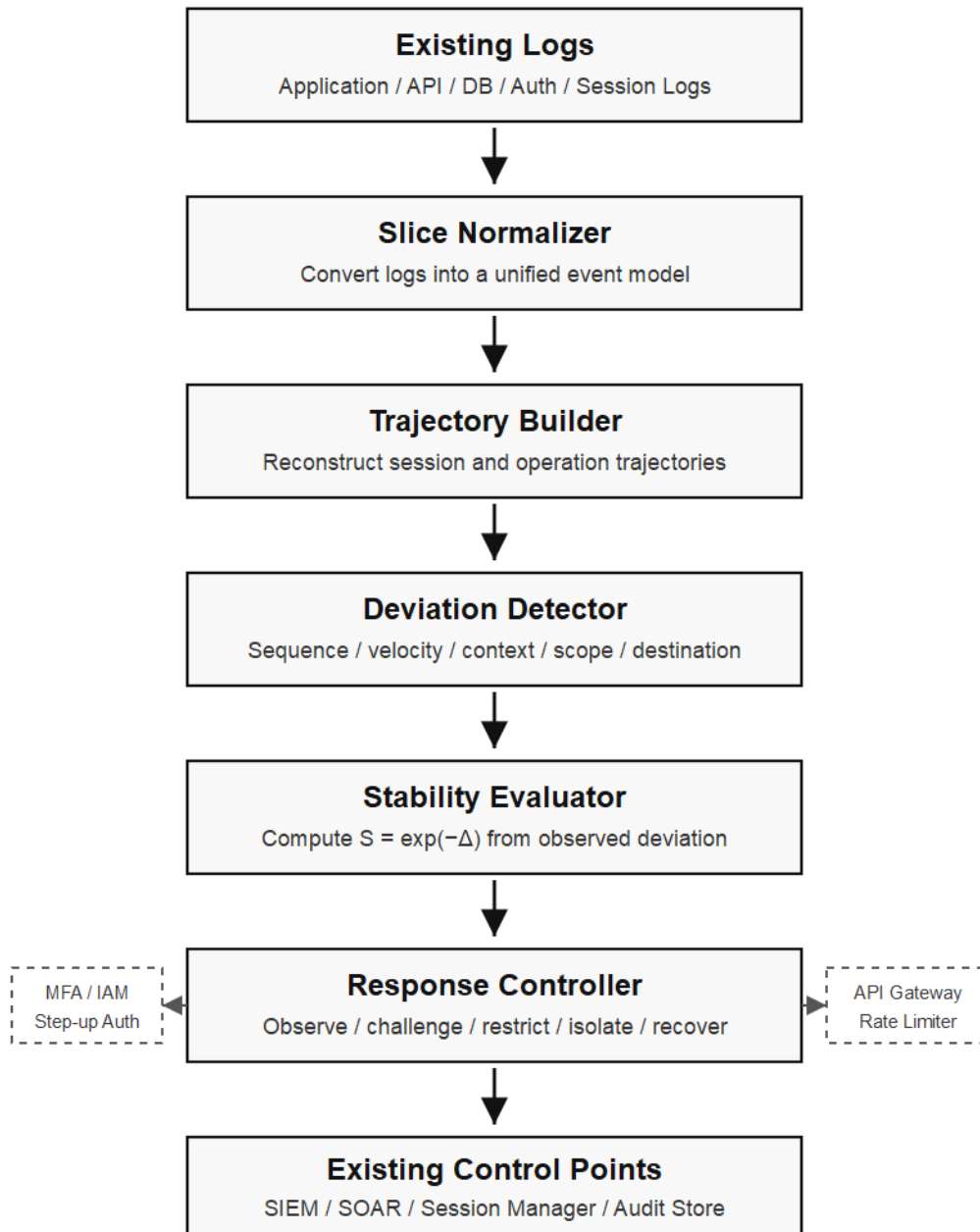


Figure 2: Existing-system overlay architecture for trajectory-based monitoring.

8. Proof of Concept

A minimal proof-of-concept was implemented under:

examples/trajectory_vulnerability_response/

The PoC uses synthetic JSONL access logs and contains three sessions:

1. normal session,
2. suspicious session,
3. attack-like session.

The PoC follows the flow:

Logs -> Slice -> Trajectory -> Deviation -> Stability -> Response

8.1 Scenario

The normal session is:

login -> view_dashboard -> search -> view_record -> logout

The suspicious session is:

login -> api_access -> export

The attack-like session is:

login -> api_access -> bulk_download -> external_transfer

All events are marked as success. This is intentional. The PoC demonstrates that successful events may still form an unstable trajectory.

8.2 Deviation Categories

The PoC evaluates the following deviations:

Deviation	Meaning
sequence	unusual event order
velocity	too many events in a short time window
context	unknown device, unusual network, unusual region, or unusual time
scope	access beyond normal resource scope
volume	unusual data volume
permission	role/resource mismatch or privilege-boundary anomaly
destination	external transfer or unknown destination

8.3 Stability Calculation

The PoC uses a deliberately simple model:

$$\Delta(T_s) = w_{\text{seq}}\Delta_{\text{seq}} + w_{\text{vel}}\Delta_{\text{vel}} + w_{\text{ctx}}\Delta_{\text{ctx}} + w_{\text{scope}}\Delta_{\text{scope}} + w_{\text{vol}}\Delta_{\text{vol}} + w_{\text{perm}}\Delta_{\text{perm}} + w_{\text{dest}}\Delta_{\text{dest}}$$

Stability is computed as:

$$S(T_s) = \exp(-\Delta(T_s))$$

This is an implementation model for the PoC. It is not a redefinition of Gyro Logic.

8.4 Response Mapping

The PoC maps stability to staged response states:

Stability	Auth State	Response
$S \geq 0.85$	AUTH_STABLE	continue
$S \geq 0.65$	RECONVERGING	observe
$S \geq 0.45$	AUTH_CHALLENGE	challenge
$S \geq 0.25$	AUTH_RESTRICTED	restrict
$S < 0.25$	AUTH_ISOLATED	isolate

8.5 Results

The PoC produced the following results:

Session	Delta	Stability	Auth State	Response
normal	0.000	1.0000	AUTH_STABLE	continue
suspicious	0.475	0.6219	AUTH_CHALLENGE	challenge
attack-like	2.705	0.0669	AUTH_ISOLATED	isolate

The stability results are summarized in Figure 3.

The normal session remained stable. The suspicious session triggered a challenge response, requiring MFA and device verification. The attack-like session triggered isolation, including session isolation, token revocation, external transfer blocking, audit log preservation, and administrator notification.

This demonstrates the main claim of the PoC:

Successful events can still form an unstable trajectory.

Figure 3. PoC Stability Results

Successful events can still form unstable trajectories

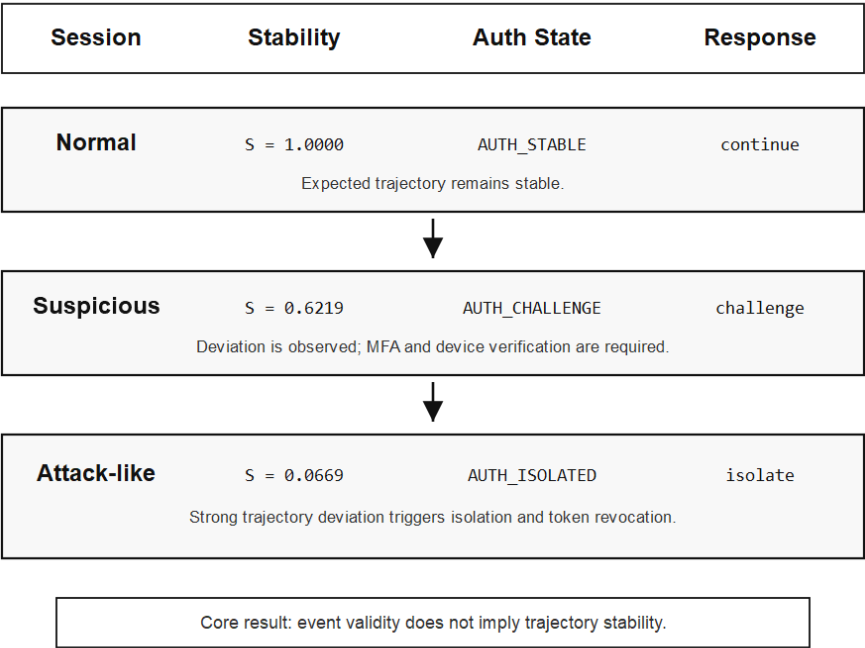


Figure 3: PoC stability results for normal, suspicious, and attack-like sessions.

9. Discussion

The proposed model is related to existing security monitoring concepts such as continuous authentication, risk-based authentication, adaptive authentication, behavioral biometrics, intrusion detection, SIEM, EDR, user/entity behavior analytics, Zero Trust Architecture, and threat-modeling frameworks. However, the emphasis here is different. The model does not only classify isolated events as normal or abnormal. It reconstructs event sequences as trajectories and evaluates whether they remain stable within identity, permission, resource, and operational context.

Compared with continuous authentication and behavioral biometrics, the proposed model does not primarily attempt to determine whether the current operator is the legitimate user through biometric or behavioral classification. Compared with risk-based authentication, it does not only compute a risk score for a login or transaction and then trigger step-up authentication. Compared with general anomaly detection, it does not treat abnormality as a property of a single event alone. Instead, it evaluates whether a sequence of individually valid events forms a stable or impermissible trajectory.

The model is also compatible with existing systems. It can begin as passive monitoring using logs, then progress toward alerting, step-up authentication, dynamic restriction, and session isolation. This framing is consistent with established cybersecurity practice in which prevention, detection, response, and recovery are treated as complementary functions rather than mutually exclusive controls.

This staged approach is important because deviation does not always mean attack. A user may travel, change devices, perform legitimate bulk work, or respond to an emergency. Therefore, Stability Response should not always be immediate termination. It should be a

staged operator-side response.

10. Ethical and Security Considerations

This work does not provide exploit code, vulnerability discovery procedures, or operational attack instructions. The PoC uses synthetic logs only and is intended to demonstrate post-login trajectory monitoring and staged response.

The proposed model should be understood as a complementary monitoring and response layer. It does not replace secure coding, patch management, access control, vulnerability management, incident response, or existing security tools.

11. Limitations

This paper presents a conceptual model and a minimal PoC. It has the following limitations:

- the PoC uses synthetic logs,
- no real-world dataset evaluation is performed,
- thresholds and weights are illustrative,
- no production intrusion detection claim is made,
- false positives require operational tuning,
- response actions require integration with real control points,
- the model does not replace patching, secure coding, access control, or existing security tools.

The purpose of the PoC is not to reproduce an exploit. It is to demonstrate that individually successful events can form an unstable trajectory and that such deviation can be mapped to staged response.

12. Conclusion

This paper proposed Trajectory-Based Vulnerability Response as an applied GyroAuth model for post-login security monitoring. The model begins from the observation that a vulnerability exploit may consist of individually valid events while forming an impermissible operational trajectory.

The proposed framework consists of three layers:

Boundary / Negative Definition = prevention

Trajectory Detection = post-entry abnormal transition detection

Stability Response = containment, re-authentication, isolation, and recovery

The minimal PoC demonstrates that successful events can be reconstructed as session trajectories, evaluated for deviation, converted into stability values, and mapped to staged response actions.

The central conclusion is:

Event validity is not trajectory stability.

For AI-assisted vulnerability discovery and increasingly complex systems, vulnerability response should evaluate not only whether events are allowed, but also whether the trajectory formed by those events remains stable.

Use of AI-assisted tools

AI-assisted tools were used for drafting support, language refinement, structural organization, and consistency checking. The author reviewed, edited, and takes full responsibility for the content, claims, references, and final manuscript.

Conflict of Interest

The author declares no competing interests.

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